

MANASSAS, VA, USA

WMO: 724036

Lat: 38.721N

Lon: 77.515W

Elev: 192

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03710

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	15.9	-1.9	5.0	16.3	2.5	6.3	20.9	25.8	30.3	23.6	31.0	3.0	330	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	93.0	74.8	90.8	74.4	88.5	73.5	78.0	88.6	76.7	86.6	75.4	84.7	7.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	132.1	83.0	73.4	125.1	81.3	72.5	121.0	80.4	41.4	88.9	40.1	86.2	38.9	85.1	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.4	18.6	16.3	2.4	96.4	6.1	2.6	-1.9	98.2	-5.5	99.7	-8.9	101.2	-13.3	103.0		
			1.0	79.9	7.7	1.7	-4.6	81.2	-9.1	82.2	-13.5	83.2	-19.1	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	33.0	37.1	43.3	55.1	63.1	72.1	76.2	74.9	68.5	56.6	44.8	36.0
	DBStd	17.21	10.58	9.42	8.92	8.73	7.88	6.08	4.67	4.95	6.49	8.31	8.32	8.51
	HDD50	1878	542	373	248	43	2	0	0	0	0	26	201	443
	HDD65	4752	994	781	674	318	135	14	1	1	40	287	608	899
	CDD50	3755	13	12	41	195	407	663	814	773	554	230	44	9
	CDD65	1153	0	0	2	20	75	227	349	309	144	26	1	0
	CDH74	11167	0	5	40	313	799	2185	3431	2816	1312	251	12	3
	CDH80	4368	0	0	9	88	262	847	1513	1147	444	57	1	0
(14) Wind	WSAvg	5.6	6.4	6.1	7.5	7.2	5.6	5.1	4.4	4.1	4.8	5.0	5.4	5.5
(15) Precipitation	PrecAvg	41.3	3.0	2.2	3.8	3.3	4.1	4.0	4.1	3.3	4.1	3.1	3.2	3.2
	PrecMax	56.4	6.6	6.5	9.0	5.0	8.2	8.4	7.1	7.3	10.2	8.1	6.2	5.7
	PrecMin	29.4	1.0	0.6	0.0	1.6	1.4	1.4	1.3	1.2	0.0	0.1	0.9	1.6
	PrecStd	6.2	1.6	1.2	2.0	1.1	1.9	1.8	1.6	1.4	2.6	2.0	1.6	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.2	72.4	79.4	87.6	90.9	93.4	97.1	95.3	92.9	85.0	75.1	67.9
		MCWB	59.9	59.8	60.9	67.8	72.3	75.3	76.7	75.5	72.4	72.0	62.6	57.8
	2%	DB	63.3	63.6	71.9	82.2	86.4	90.5	93.1	92.5	89.0	81.0	69.9	61.1
		MCWB	56.6	52.3	56.1	64.2	70.6	74.2	75.5	75.0	71.6	67.2	60.1	53.1
	5%	DB	56.8	57.2	65.7	77.0	82.3	88.2	90.7	90.0	84.7	75.4	64.3	55.0
		MCWB	50.4	47.1	53.8	62.0	68.6	73.4	75.2	74.4	70.3	65.2	56.6	48.2
	10%	DB	48.2	52.3	60.8	72.3	78.7	85.0	88.3	87.0	81.4	72.2	60.8	50.1
		MCWB	42.6	45.0	49.9	59.4	65.9	71.9	74.2	73.4	67.7	63.4	52.6	43.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	62.3	62.9	70.4	76.0	77.9	80.3	79.3	77.8	74.7	65.4	60.3
		MCDB	65.5	70.4	75.7	80.4	86.9	89.5	92.0	90.7	86.0	81.6	70.6	65.3
	2%	WB	57.8	54.4	58.9	67.2	72.8	76.3	78.3	77.4	74.9	71.8	62.6	54.8
		MCDB	62.6	62.2	69.0	78.1	82.6	87.0	89.4	87.4	82.3	76.6	68.0	59.9
	5%	WB	50.8	49.7	54.9	64.2	70.4	74.9	77.0	76.1	72.9	68.5	58.1	49.3
		MCDB	56.5	54.6	63.5	73.9	79.7	84.8	87.0	85.6	80.5	73.9	63.4	54.9
	10%	WB	43.2	45.5	50.5	61.0	68.2	73.4	75.5	74.7	71.0	64.2	53.4	44.0
		MCDB	47.8	51.2	59.7	70.8	76.1	82.8	84.8	83.4	78.4	70.3	58.8	49.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.7	22.2	24.5	22.6	21.3	20.8	20.7	22.2	23.6	22.3	19.5
		MCDBR	27.7	29.6	32.3	32.5	27.9	24.8	24.2	24.7	27.3	27.5	29.1	27.2
		MCWBR	21.3	20.7	19.5	17.1	14.5	11.3	10.0	9.9	12.5	16.2	19.9	19.9
	5% WB	MCDBR	24.5	24.4	27.7	26.7	23.7	21.9	21.3	20.2	20.8	20.7	22.8	24.9
		MCWBR	20.2	18.8	18.8	15.6	13.0	10.9	9.5	9.2	11.0	13.2	18.0	19.5
(40) Clear-Sky Solar Irradiance	taub		0.304	0.320	0.355	0.399	0.438	0.498	0.515	0.513	0.425	0.359	0.322	0.308
	taud		2.489	2.422	2.388	2.287	2.205	2.077	2.052	2.037	2.261	2.448	2.511	2.514
	Ebn at Noon		278	287	286	279	269	252	246	243	259	269	269	268
	Edh at Noon		25	30	35	41	45	52	53	52	39	30	24	23
(44) All-Sky Solar Radiation	RadAvg		665	928	1235	1546	1745	1922	1878	1680	1376	1039	756	585
	RadStd		59	104	83	115	183	145	100	83	166	116	63	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+137

Nomenclature: See separate page